

The structure looks something like an inflatable backyard bounce house for children, but it is far more sophisticated. It is insulated and heated, has power and is pressurized. On the moon, it could be used to evaluate materials, lightweight structure technologies, astronaut interfaces, dust mitigation techniques, and function with robotics and other lunar surface equipment.

Of course, this is still a pie-in-the-sky project, much like the programmes of going to Mars.

Mars500

NASA, the Russian Federal Space Agency and the China National Space Administration have all announced their intentions of going to Mars – some as early as 2030 and others as late as 2060. So far, apart from landing a few robotic probes on the Red Planet, all of the preparations are in the preliminary stage. The most ambitious and publicised part of the projects, though, is the Mars500.

The purpose of the Mars500 study is to gather data, knowledge and experience to help prepare for a real mission to Mars. Obviously there has not been effect of weightlessness, but the study also helps to determine key psychological and physiological effects of being in such an enclosed environment for such an extended period of time.

In order to simulate a mission to Mars, six candidates (three Russian, two European and one Chinese) have been sealed in an isolation chamber since June 2010. This group was chosen to encompass working experience in many fields, including medicine, engineering, biology and computer engineering.

Part of the chamber simulates the spacecraft that would transport them on their journey to and from Mars and another part will simulate the landing module that would transfer them to and from the Martian surface.

Following the completion of an initial 105-day isolation period in 2009, a full 520-day study was started in June 2010, which ended in November 2011.

Once sealed into the chamber, the candidates had only personal contact with each other plus voice contact with a simulated control centre and family and friends as would normally happen in a human spaceflight mission. A 20-minute



The Mars500 Crew. Standing: (left to right) Sukhrob Kamolov, Alexandr Smoleevskiy, Diego Urbina. Sitting (left to right) Yue Wang, Alexey Sitev, Romain Charles